

BOGUCHANY, Russian Federation

WMO: 292820

Lat: **58.3838N**Lon: **97.4507E**Elev: **134**StdP: **99.73**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.5	(c) -39.4	(d) -44.3	(e) 0.0	(f) -42.0	(g) -41.5	(h) 0.1	(i) -39.2	(j) 6.7	(k) -12.0	(l) 5.9	(m) -12.4	(n) 0.1	(o) 260	(p) 0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	29.6	18.5	27.7	17.8	25.7	16.9	20.1	26.4	19.0	25.2	18.0	23.9	2.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.1	22.6	16.7	12.1	21.5	15.7	11.4	20.7	58.4	26.5	54.9	25.2	51.7	23.9	
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Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.1	(b) 5.3	(c) 4.6	(e) -42.6	(f) 33.1	(g) 3.7	(h) 1.5	(i) -45.2	(j) 34.2	(k) -47.4	(l) 35.1	(m) -49.5	(n) 35.9	(o) -52.2	(p) 37.0		
(a) 6.1	(b) 5.3	(c) 4.6	(e) -42.4	(f) 21.7	(g) 3.4	(h) 1.1	(i) -44.8	(j) 22.5	(k) -46.8	(l) 23.1	(m) -48.7	(n) 23.7	(o) -51.1	(p) 24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.0	-23.1	-19.0	-8.4	1.3	8.5	16.7	19.0	15.8	8.3	0.2	-11.9	-20.6		
	DBStd	16.11	10.16	8.87	7.57	4.79	4.84	3.96	3.09	3.47	4.22	5.37	9.37	10.38		
	HDD10.0	4758	1025	811	572	262	90	3	0	2	81	305	658	950		
	HDD18.3	7152	1283	1044	830	510	307	76	29	92	301	563	908	1208		
	CDD10.0	740	0	0	0	1	42	203	279	183	31	1	0	0		
	CDD18.3	93	0	0	0	0	1	27	51	14	0	0	0	0		
	CDH23.3	1014	0	0	0	0	37	344	475	152	6	0	0	0		
	CDH26.7	269	0	0	0	0	7	98	132	32	1	0	0	0		
	WSAvg	1.7	1.3	1.4	1.8	2.2	2.1	1.6	1.4	1.4	1.7	2.1	2.0	1.5		
	PrecAvg	343	16	12	11	18	32	42	52	51	44	26	25	20		
(15) Precipitation	PrecMax	498	36	24	26	44	78	135	153	121	105	65	41	38		
	PrecMin	220	5	4	1	3	5	0	6	5	13	5	7	4		
	PrecStd	68	7	5	6	8	16	27	32	23	20	12	8	8		
	PrecStd	68	7	5	6	8	16	27	32	23	20	12	8	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	1.6	9.4	18.1	27.2	31.9	32.2	30.1	23.9	13.6	7.0	0.6		
		MCWB	-2.0	-1.0	3.6	8.3	14.1	18.1	20.2	19.3	15.0	8.0	4.3	-1.0		
	2%	DB	-2.9	-0.4	5.9	14.5	23.5	29.4	29.8	27.1	19.9	10.8	4.0	-1.4		
		MCWB	-4.2	-2.4	1.5	7.0	12.7	17.5	19.1	18.0	13.2	7.2	1.9	-2.7		
	5%	DB	-5.5	-2.6	3.7	11.5	20.5	27.3	28.0	24.8	17.1	8.9	2.5	-3.5		
		MCWB	-6.6	-4.2	0.3	5.6	11.5	16.7	18.7	17.0	12.0	6.0	0.8	-4.6		
	10%	DB	-8.4	-5.7	1.9	8.9	17.7	25.0	26.1	22.7	14.7	7.0	0.6	-6.0		
		MCWB	-9.3	-6.9	-0.8	4.0	10.2	16.1	17.8	16.0	11.0	4.7	-0.7	-7.0		
	0.4%	WB	-1.9	-0.5	4.0	9.7	15.6	20.4	22.0	20.4	16.1	9.3	4.8	-1.0		
		MCDB	-0.6	1.5	8.1	15.7	23.6	27.0	29.3	27.2	21.4	11.6	6.7	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-4.1	-2.2	1.9	7.5	13.9	19.0	20.6	18.9	14.1	7.8	2.3	-2.6		
		MCDB	-2.9	-0.5	5.1	13.5	21.5	25.6	26.6	25.5	18.5	10.1	3.8	-1.5		
	5%	WB	-6.6	-4.2	0.6	5.9	12.3	18.0	19.6	17.7	12.7	6.2	0.8	-4.6		
		MCDB	-5.5	-2.5	3.6	10.9	19.3	25.1	25.7	23.0	16.1	8.6	2.3	-3.6		
	10%	WB	-9.2	-6.9	-0.9	4.3	10.7	17.0	18.7	16.7	11.4	4.9	-0.8	-7.0		
		MCDB	-8.3	-5.6	2.0	8.6	16.9	23.5	24.4	21.3	14.4	7.0	0.7	-6.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	10.5	12.4	10.1	11.5	11.9	11.1	10.2	8.5	6.2	6.2	7.3		
		MCDBR	8.0	9.9	12.1	14.7	16.6	16.6	14.6	14.4	13.3	9.4	5.7	7.9		
		MCWBR	7.4	8.7	8.7	8.4	7.9	6.3	5.6	6.2	6.7	6.2	4.8	7.3		
		MCDBR	8.1	9.7	11.1	13.3	15.1	13.7	11.9	12.4	11.0	8.4	5.7	8.2		
	5% WB	MCWBR	7.5	8.7	8.3	8.0	7.8	6.0	5.1	5.8	6.1	5.8	4.9	7.6		
		MCWBR	7.5	8.7	8.3	8.0	7.8	6.0	5.1	5.8	6.1	5.8	4.9	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.308	0.340	0.373	0.393	0.431	0.456	0.418	0.361	0.336	0.291	0.264		
	taud		1.981	2.006	2.010	2.131	2.267	2.230	2.173	2.249	2.377	2.260	2.040	1.943		
	Ebn at Noon		500	683	779	825	838	806	772	776	772	663	483	384		
	Edh at Noon		58	92	122	129	121	128	133	115	86	72	54	43		
(44) All-Sky Solar Radiation	RadAvg		0.61	1.56	2.90	4.43	5.09	5.88	5.41	4.25	2.54	1.21	0.60	0.34		
	RadStd		0.08	0.13	0.20	0.24	0.36	0.51	0.36	0.28	0.35	0.11	0.09	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page